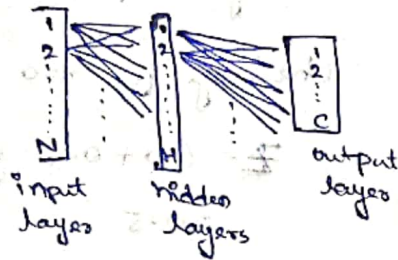


1. The given the neural networks are:-  
 N - input neurons  
 H - hidden neurons  
 C - output neurons



every connection corresponds to a weight

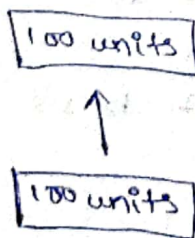
- a) if connections between input and hidden layers and between hidden and output layers

$$\text{total weight} = N \times H + H \times C$$

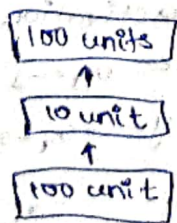
- b) if connections are between input and hidden layers, hidden layer and output layers, input and output layers then,

$$\text{total weights} = N \times H + H \times C + N \times C$$

i) Network A



Network B

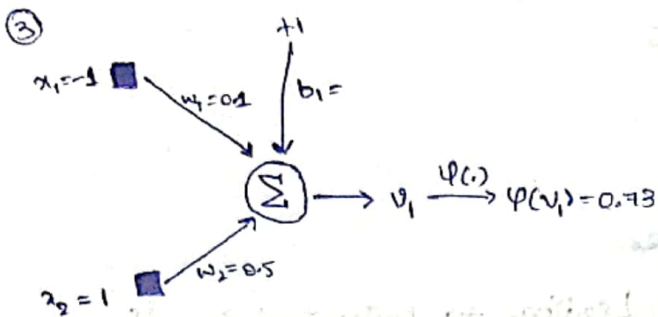


- i) Network A, being shallower, often reduces computational complexity. With fewer layers, it requires less processing power and memory.

Leading to faster training times and more straight forward implementation. This efficiency is beneficial when working with limited computational resources or when rapid prototyping is needed.

- ii) Network B, with its additional layers possesses an enhanced capacity to model complex functions. The increased depth allows the network to represent more intricate patterns and relationships.

However, it's important to note that in the absence of non-linear activation functions, the overall transformation of such a network remains linear, regardless of depth. Therefore to fully leverage the representational power of deeper networks incorporating non-linear activation functions are essential.



logistic Activation function with slope parameter  $\alpha = 2$

$$\therefore \phi(x) = \frac{1}{1 + e^{-\alpha x}} = \frac{1}{1 + e^{-2x}}$$

$$v_1 = x_1 w_1 + x_2 w_2 + b_1 \cdot 1$$

$$= -0.1 + 0.5 + 1$$

$$= 1.4$$

$$\phi(v_1) = 0.73$$

$$\Rightarrow \frac{1}{1 + e^{-2v_1}} = \frac{1}{0.73}$$

$$e^{-2v_1} = 0.3698$$

$$v_1 = 0.497311$$

$$\Rightarrow b_1 = 0.097311$$

Value of bias  $b_1 = 0.0973$

④ a) Perception for NOT Gate

We want a perception that takes a single binary input

$x \in \{0, 1\}$  and output:

$$\begin{cases} 1 & \text{if } x=0 \\ 0 & \text{if } x=1 \end{cases}$$

Input ( $x$ )      output (NOT  $x$ )

1      0

0      1

We choose our activation function as,

$$y = \text{step}(z) = \begin{cases} 0 & , z < 0 \\ 1 & , z \geq 0 \end{cases}$$

$\therefore$  Equation:-

$$y = \text{step}(wx + b), \quad z = w \cdot x + b$$

We choose weights

$$w = -1$$

$$\text{bias } b = 0.5$$

$$\therefore y = \text{step}(-x + 0.5)$$

checking for  $x=0$ ,

$$z = (-0 + 0.5) \quad y = 1 \\ = 0.5$$

$$x=1,$$

$$z = -1 + 0.5 \quad y = 0 \\ = -0.5$$

$\therefore$  Hence satisfied//

b) Perception of NAND gate:

Input  $x_1$     Input  $x_2$     Output (NAND)

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

Perception equation:-

$$y = \text{step}(w_1 x_1 + w_2 x_2 + b)$$

using step function as activation function

choose  $w_1 = -1$ ,  $w_2 = -1$ ,  $b = 1.5$

$$y = -x_1 - x_2 + 1.5$$

| $x_1$ | $x_2$ | $z = -x_1 - x_2 + 1.5$ | $y$ |
|-------|-------|------------------------|-----|
| 0     | 0     | 1.5                    | 1   |
| 0     | 1     | 0.5                    | 1   |
| 1     | 0     | 0.5                    | 1   |
| 1     | 1     | -0.5                   | 0   |

$\therefore$  Hence Satisfied//





- ⑤ Using two perceptrons to compute parity, we need to get the parity function for three binary inputs:

$$\text{Parity}(x_1, x_2, x_3) = x_1 \oplus x_2 \oplus x_3$$

This can be achieved in two stages:  
first compute  $x_1 \oplus x_2$ , then XOR the result with  $x_3$

Perceptron 1:-

$$\text{Compute } z = x_1 \oplus x_2$$

We define a small MLP to compute XOR:

→ Hidden layer:

$$-h_1 = \text{step}(0.91x_1 + 0.88x_2 - 0.43)$$

$$-h_2 = \text{step}(-0.97x_1 - 0.92x_2 + 1.6)$$

→ Output layer:

$$z = \text{step}(1.02h_1 + 1.01h_2 - 1.49)$$

Perceptron 2 :-

$$\text{Compute } y = z \oplus x_3$$

Again, using the XOR architecture:

→ Hidden layer:

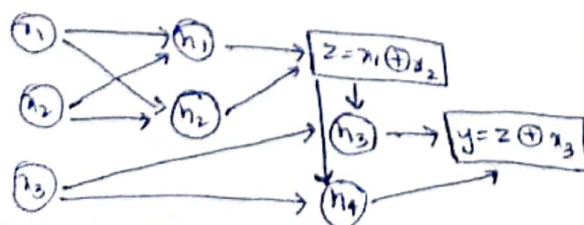
$$-h_3 = \text{step}(1.07z + 0.95x_3 - 0.51)$$

$$-h_4 = \text{step}(-1.11z - 1.03x_3 + 1.59)$$

→ Output:

$$y = \text{step}(1.03h_3 + 0.94h_4 - 1.41)$$

Final output:  $y = 1$  if the number of 1s among  $x_1, x_2, x_3$  is odd (i.e., parity is 1) else 0



- ⑥ for the MLP network:

$$\text{Input: } x \in \mathbb{R}^D$$

$$\text{Hidden layers: } H \text{ units}$$

$$\text{Weights: } w_i \in \mathbb{R}^{H \times D}$$

$$\text{Biases: } b_i \in \mathbb{R}^H$$

Pre-Activation:-

$$z_i = \sum_{j=1}^D w_{ij} x_j + b_{i1} \quad \text{for } i = 1, \dots, H$$

$$\text{Activation: } h_i = \text{ReLU}(z_i) = \max(0, z_i)$$

Output layer:-

$$\text{Weights: } w_2 \in \mathbb{R}^H$$

$$\text{Bias: } b_2 \in \mathbb{R}$$

$$\text{Output: } \hat{y} = \sum_{i=1}^H w_{2,i} h_i + b_2$$

Loss function:-

Squared error:-

$$L = \frac{1}{2} (\hat{y} - y)^2$$

Gradients:-

A. Output layer:-

$$\frac{\partial L}{\partial w_{2,i}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial w_{2,i}} = (\hat{y} - y) h_i$$

$$\frac{\partial L}{\partial \hat{y}} = (\hat{y} - y)$$

$$\frac{\partial \hat{y}}{\partial w_{2,i}} = h_i$$

$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial b_2} = (\hat{y} - y)$$

$$\frac{\partial \hat{y}}{\partial b_2} = 1$$

∴ Updated equations:-

with learning rate  $\eta$

$$w_{2,i}^{\text{new}} = w_{2,i} - \eta (\hat{y} - y) h_i$$

$$b_2^{\text{new}} = b_2 - \eta (\hat{y} - y)$$



B. Hidden layer:

for a hidden unit  $i$ , which has pre-activation  $z_i$  and activation  $h_i$ ,

$$h_i = \text{ReLU}(z_i), \text{ReLU}'(z_i)$$

$$= \begin{cases} 1 & z_i > 0 \\ 0 & z_i \leq 0 \end{cases}$$

$$\frac{\partial L}{\partial w_{1,i}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial h_i} \frac{\partial h_i}{\partial z_i} \frac{\partial z_i}{\partial w_{1,i}}$$

$$\frac{\partial \hat{y}}{\partial h_i} = w_{2,i}$$

$$\frac{\partial h_i}{\partial z_i} = \text{ReLU}'(z_i) = I(z_i > 0),$$

$I(\cdot)$  is an indicator function

$$\frac{\partial z_i}{\partial w_{1,i}} = x_j$$

$$\frac{\partial L}{\partial w_{1,i}} = (\hat{y} - y) (w_{2,i}) (I(z_i > 0) x_j)$$

$$\frac{\partial L}{\partial b_{1,i}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial h_i} \frac{\partial h_i}{\partial z_i} \frac{\partial z_i}{\partial b_{1,i}} = (\hat{y} - y) (w_{2,i}) I(z_i > 0)$$

$$\frac{\partial z_i}{\partial b_{1,i}} = -1$$

Updated equations:

$$w_{1,i}^{\text{new}} = w_{1,i} - \eta (\hat{y} - y) w_{2,i} I(z_i > 0) x_j$$

$$b_{1,i}^{\text{new}} = b_{1,i} - \eta (\hat{y} - y) w_{2,i} I(z_i > 0)$$

hence  $I(z_i > 0)$  is 1 if  $z_i > 0$   
0 o.w